

Rev	Date	Eng	Changes
AA	2026-10-07	NB	First release, derived from Sofar mote 000639-AB

nereus_Pi_shield_ADIN2111

Variant: Standard

P/N: PCB-000001-AA

Index

- 1 Cover
- 2 Top-Level
- 3 ADIN2111 (Single-Pair Ethernet)
- 4 Load Switch (payload port)
- 5 PoDL (Power over Data Line)
- 6 Bus Power Monitor
- 7 Power (buck converters)

Pi shield renders: TBD (after layout)

I2c Address Table

Device	I2C Address
Power Monitor	0x41

DIMENSIONS: TBD (after layout)

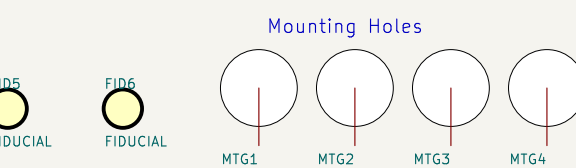
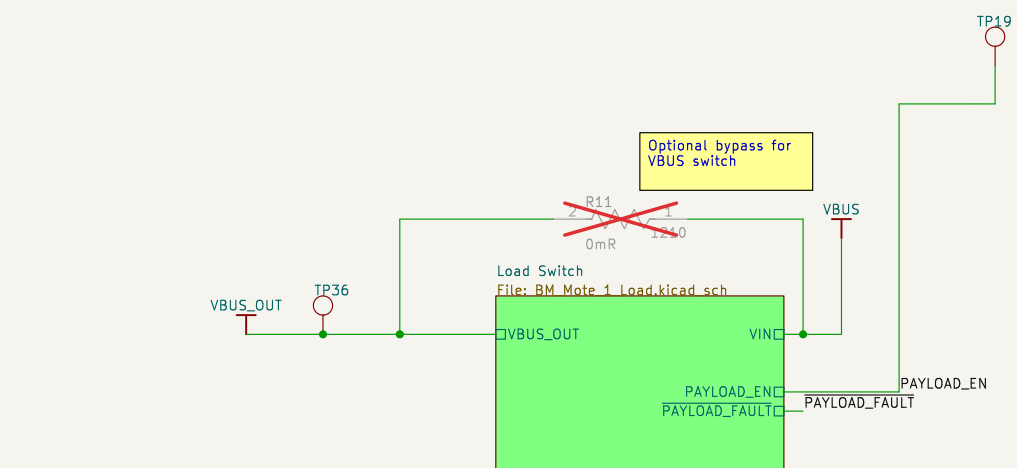
Top-Level Schematic
File: BM Mote 1 Master.kicad_sch



The Bristlemouth Mote Board is licensed under a Creative Commons Attribution-NoDerivatives 4.0 International License. The Bristlemouth trademarks and the Bristlemouth logo are registered and unregistered marks belonging to Sofar Ocean Technologies, Inc., and are not included under the Creative Commons license.
<https://creativecommons.org/licenses/by-nd/4.0/legalcode>

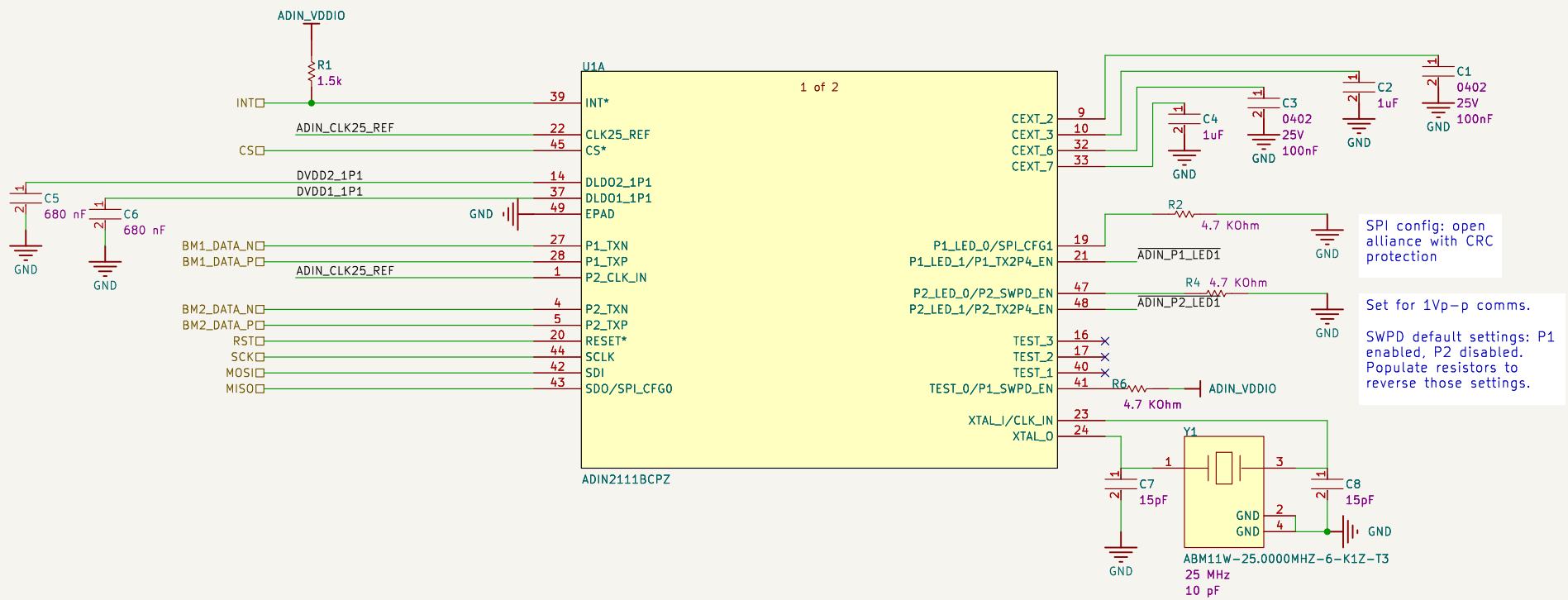


Derived from Sofar Bristlemouth mote 000639-AB Author: Nick Buemond nereus_Pi_shield_ADIN2111 · PCB-000001-AA			
Sheet: / File: nereus_Pi_shield_ADIN2111.kicad_sch			
Title: Cover Page			
Size: A3	Date: 2026-10-07		Rev: AA
KiCad E.D.A. 9.0.6		Id: 1/7	



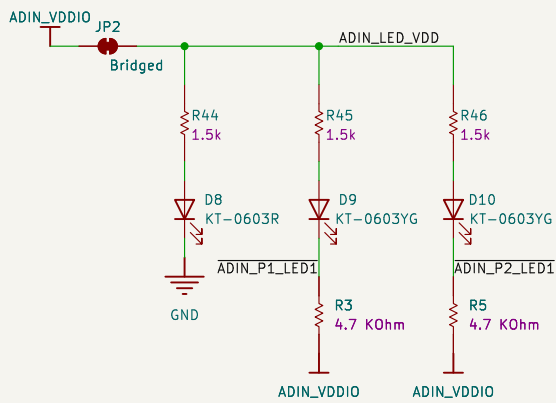
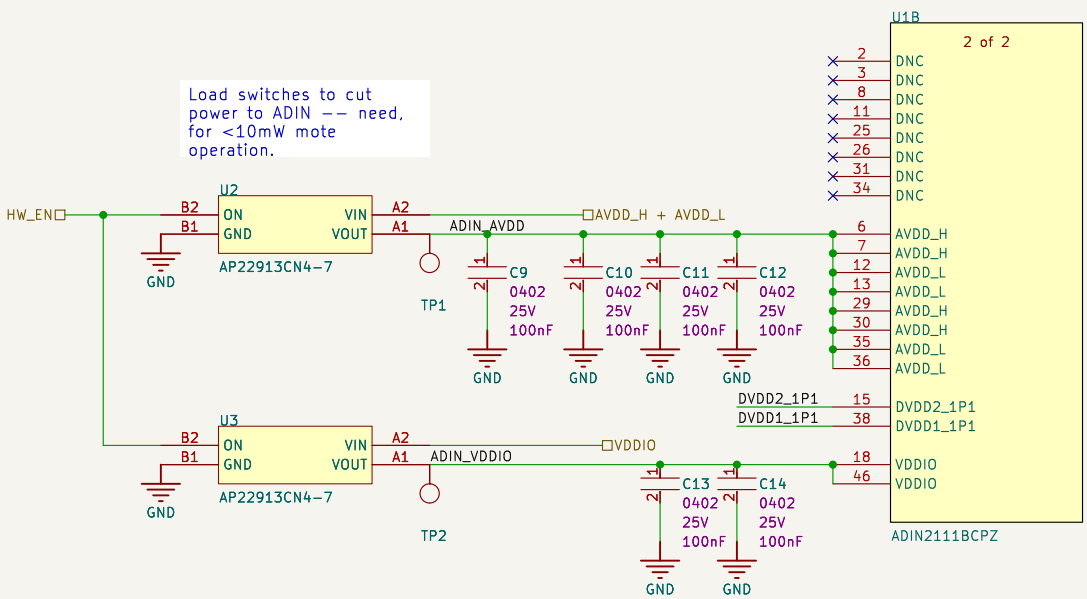
Id: 2/7

ADIN2111 Single-Pair Ethernet



ADIN2111 status LEDs: power, port 1 / port 2 link-activity

ADIN2111 supplies: U2 (AVDD) and U3 (VDDIO) switched on by ADIN_PWR



ADIN status LEDs (D25): D8 red = ADIN powered (ADIN_VDDIO on). D9 / D10 yellow-green = port 1 / port 2 link/activity. ADIN pins 21 / 48, active low. Off until the Pi enables each port's LED_1 over SPI; port 2 is also asleep until woken. R3 / R5 keep the TX2P4_EN straps high (1.0 V p-p). Cut JP2 to disable all three LEDs.

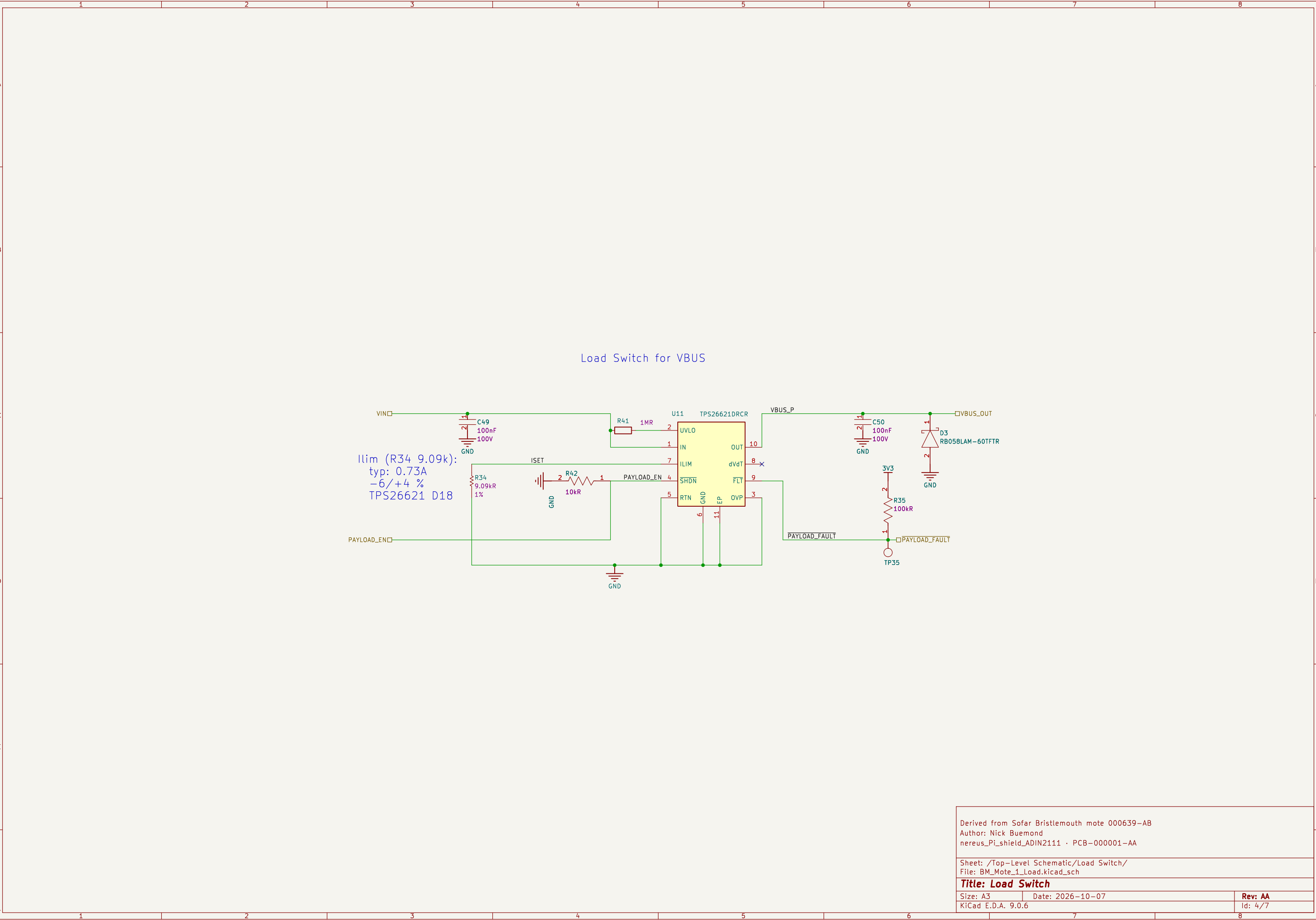
Derived from Sofar Bristlemouth mote 000639-AB
Author: Nick Buemond
nereus_Pi_shield_ADIN2111 · PCB-000001-AA

Sheet: /Top-Level Schematic/ADIN/
File: BM_Mote_1_ADIN2111.kicad_sch

Title: ADIN2111

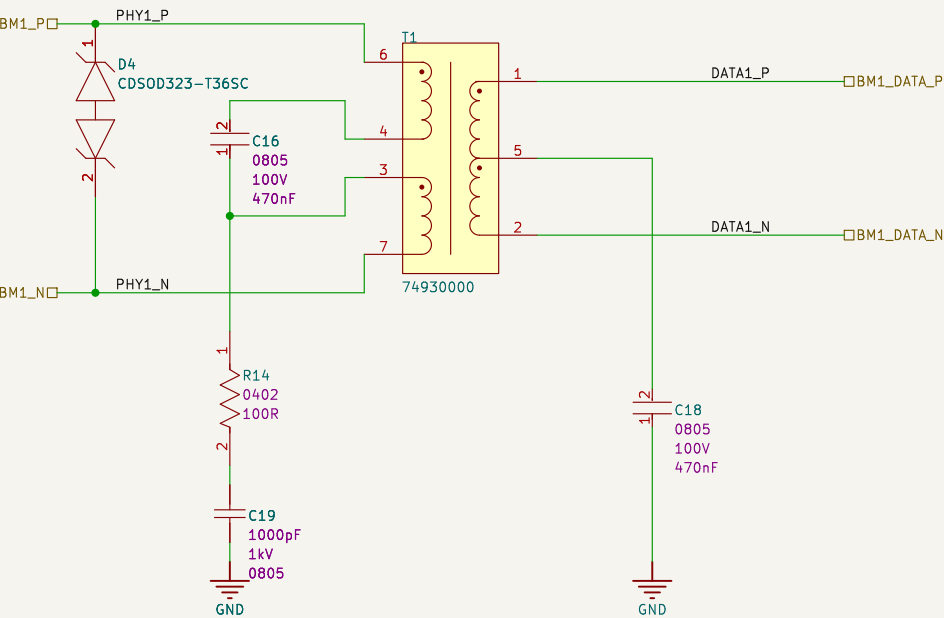
Size: A3 Date: 2026-10-07
KiCad E.D.A. 9.0.6

Rev: AA
Id: 3/7

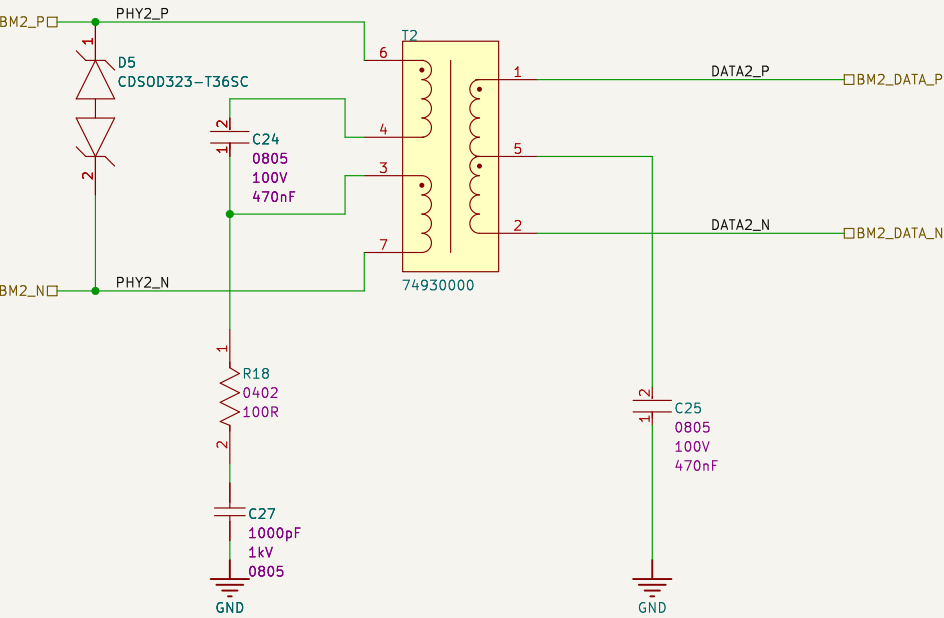


Power Over Dataline

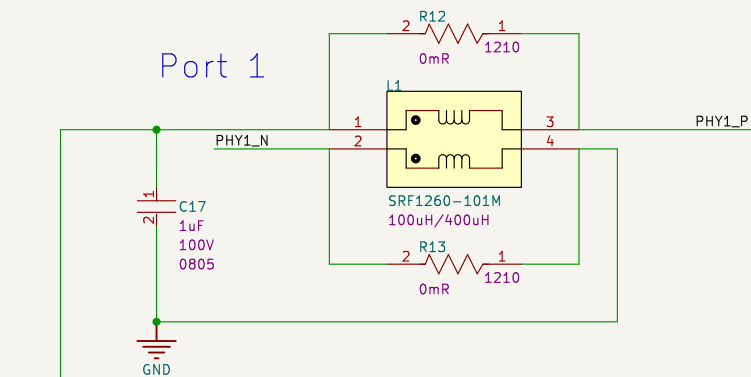
Coupling Network – Port 1



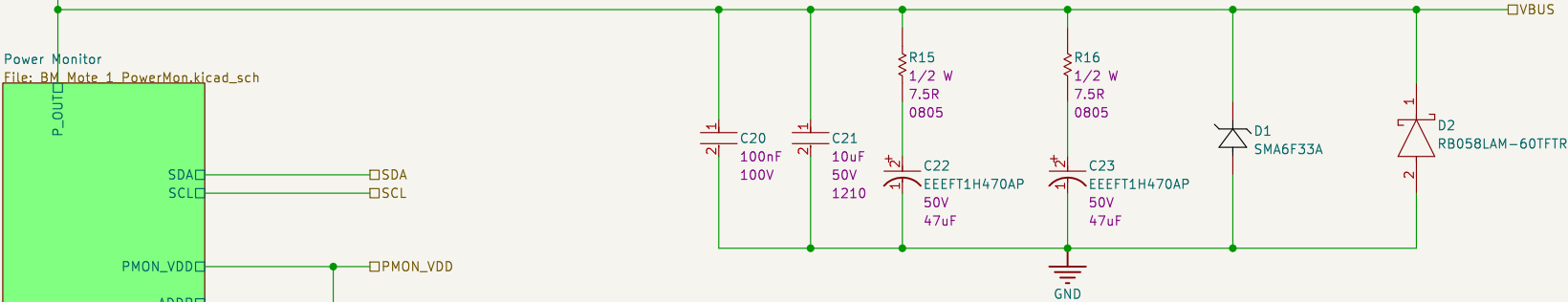
Coupling Network – Port 2



Port 1

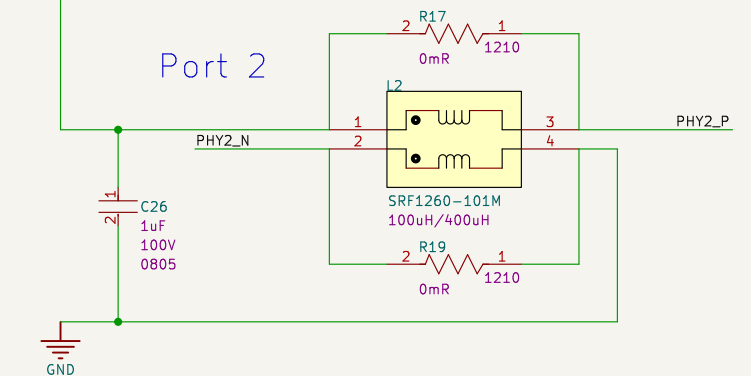


DC Connection + Circuit protections



Snubber resistors must be at least 1/2 W, to support hot-plugging.

Port 2



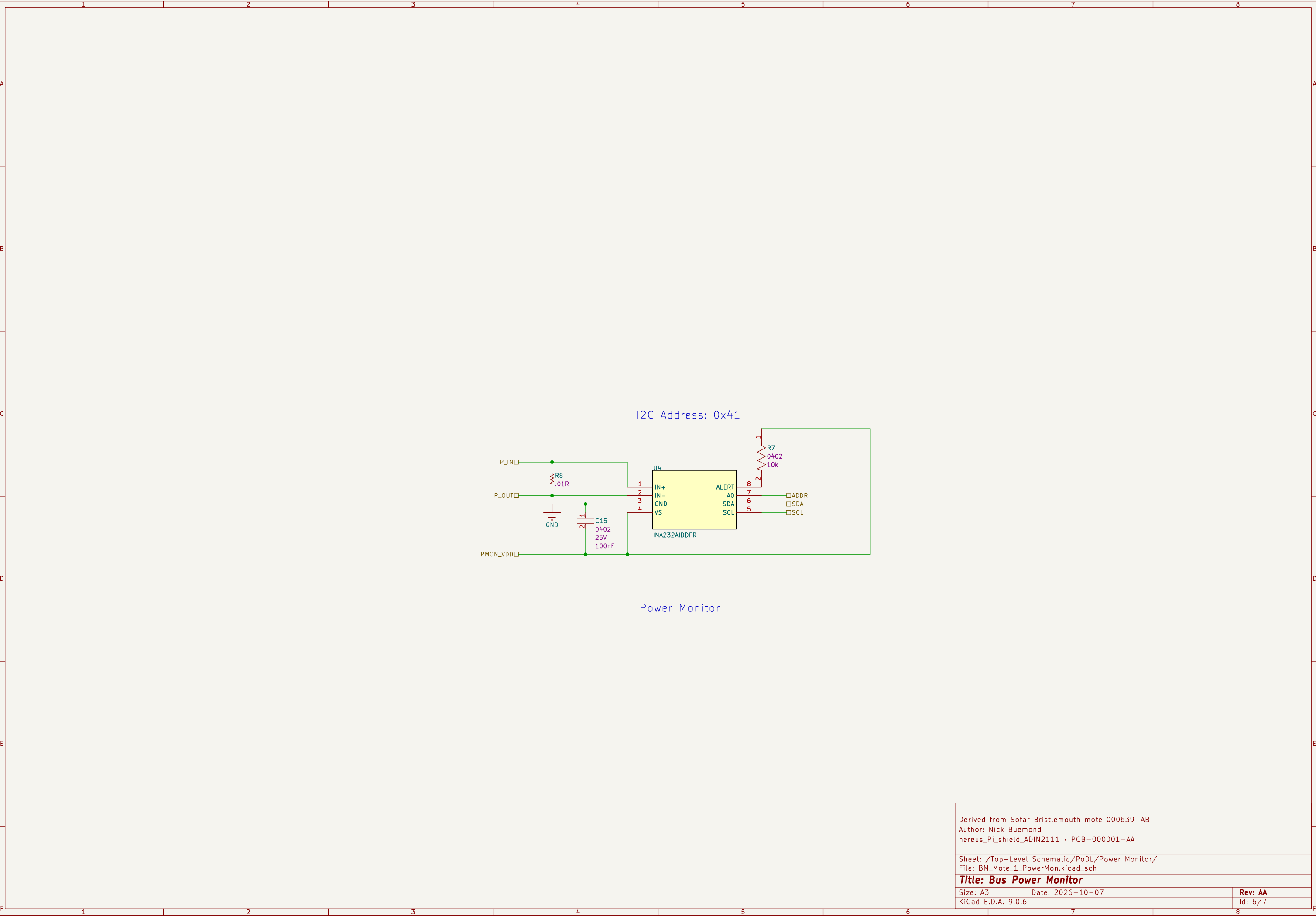
Derived from Sofar Bristlemouth mote 000639-AB
Author: Nick Buemond
nereus_Pi_shield_ADIN2111 · PCB-000001-AA

Sheet: /Top-Level Schematic/PoDL/
File: BM_Mote_1_PoDL.kicad_sch

Title: PoDL (Power over Data Line)

Size: A3
Date: 2026-10-07
KiCad E.D.A. 9.0.6

Rev: AA
Id: 5/7



Derived from Sofar Bristlemouth mote 000639-AB
Author: Nick Buemond
nereus_Pi_shield_ADIN2111 · PCB-000001-AA

Sheet: /Top-Level Schematic/PoDL/Power Monitor/
File: BM_Mote_1_PowerMon.kicad_sch

Title: Bus Power Monitor

Size: A3
KiCad E.D.A. 9.0.6

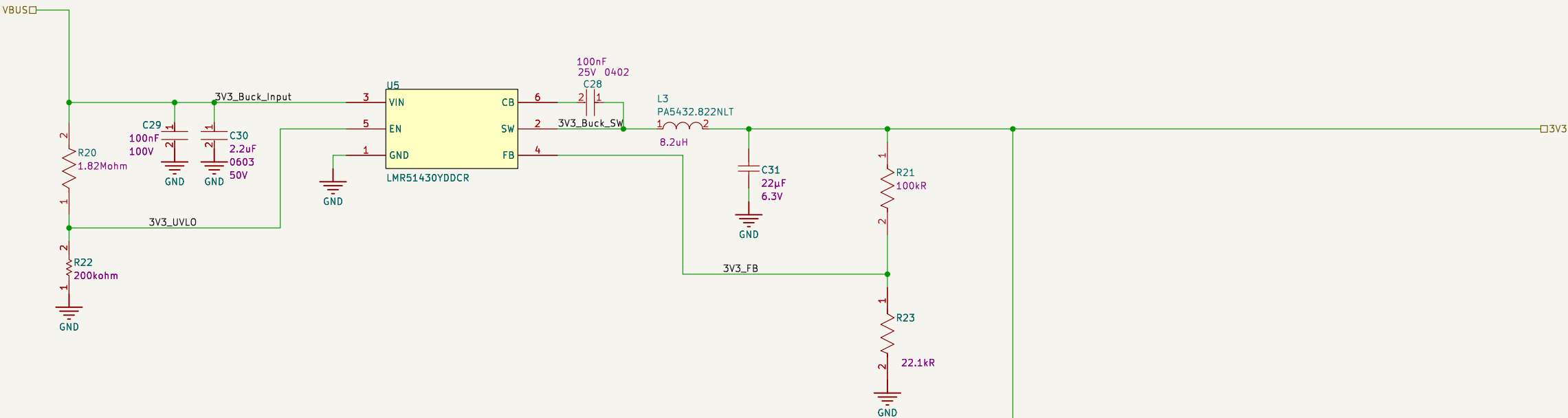
Date: 2026-10-07

Rev: AA

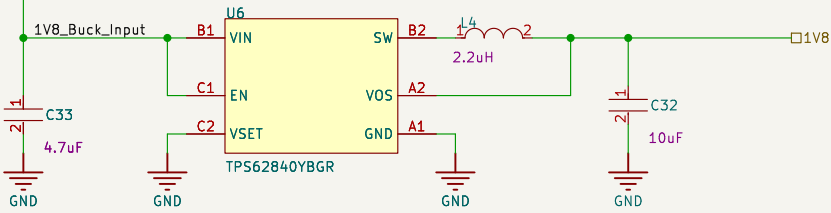
Id: 6/7

Buck Converters – 3V3, 1V8, 5V

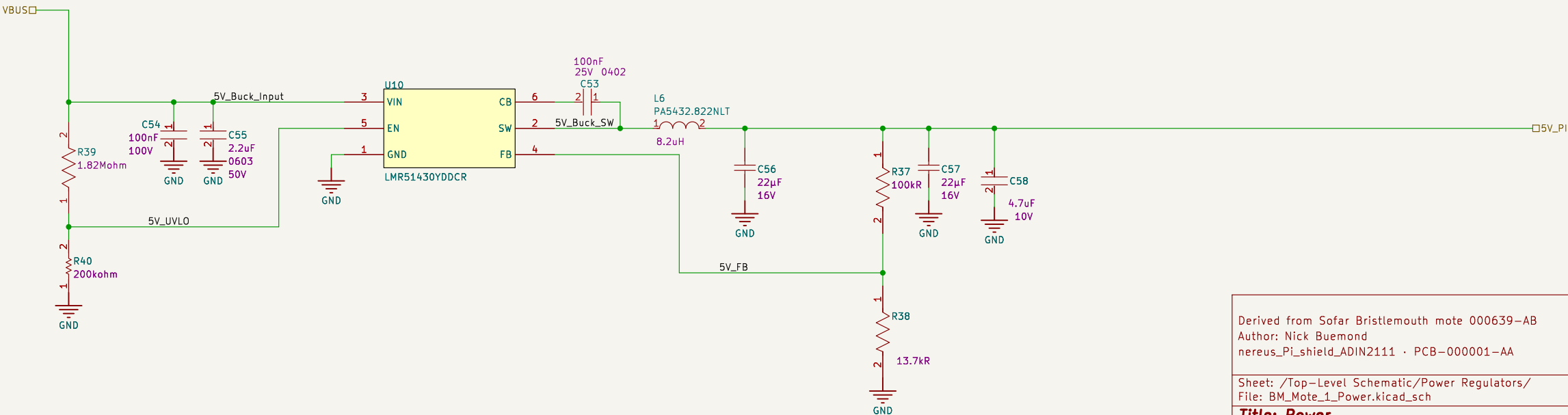
3V3 buck converter: U5 LMR51430, VBUS → 3V3



1V8 buck converter: U6 TPS62840, 3V3 → 1V8 (ADIN AVDD)



5V buck converter for the Pi: U10 LMR51430, VBUS → 5V_PI



Derived from Sofar Bristlemouth mote 000639-AB
Author: Nick Buemond
nereus_Pi_shield_ADIN2111 · PCB-000001-AA

Sheet: /Top-Level Schematic/Power Regulators/
File: BM_Mote_1_Power.kicad_sch

Title: Power

Size: A3 Date: 2026-10-07
KiCad E.D.A. 9.0.6

Rev: AA
Id: 7/7